BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 198 · Linked lists and pointers

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Push 2, 5, 8 then pop. What remains bottom-to-top?

Q2. What is the first index of a C# array?

Q3. What is a base case?

Q4. Queue A,B,C: dequeue once and enqueue D. What remains?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

node — A list element containing data and a link.

next pointer — Reference or index of the next node.

head — Reference to the first list node.

null sentinel — Marker such as −1 in an index implementation meaning no next node.

traversal — Visit nodes by following successive links.

New learning

Follow references — head = 0 means the first node has value 10. next[0] initially points at node 1 (20).

Prepare the new node — The new node at index 3 stores 15 and next[3] = 1, preserving the rest of the list.

Change the existing link — Set next[0] = 3, producing the chain 10 → 15 → 20 → 30.

Worked example

Insert value 7 between 4 and 9 in the linked list 4 → 9 → 12.

1. Predict — Create node 7 and set its next link to the existing node 9.

2. Confirm and explain — Point node 4 at node 7. The chain becomes 4 → 7 → 9 → 12.

We Do · Practise with guidance

Insert 6 after 2 in 2 → 8 → 11 without losing 8.

```
// Pointer pseudocode: node6 already exists.
node6.next = node2.next; // preserve node8
node2.next = node6;
```

Predict

First link new node 6 to node 8.

Confirm and explain

Then link node 2 to node 6, giving 2 → 6 → 8 → 11.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

In linked list 5 → 10 → 15, what follows node 10?

Question 2

A node with value 12 must be inserted between 7 and 20. State the two link changes.

Question 3

Why does inserting into a linked list avoid shifting array elements but may still take O(n) time?

Exit Ticket · Answer before checking

A singly linked list always supports direct O(1) access to its twentieth element.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 2, 5. The top 8 is removed.

Q2. 0.

Q3. The stopping condition in recursion.

Q4. B,C,D front-to-rear.

You Do pages · Independent questions

Q1. Node 15.

Q2. Set new12.next to node20, then set node7.next to new12.

Q3. Once the insertion node is found, links can be updated in O(1), but finding the position usually needs a traversal that takes O(n).

Exit Ticket · Model answer

False. Traversal from the head is generally required.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: